ME 364 SYSTEMATIC ENGINERING DESIGN
Fall 2025 SYLLABUS

Course Information

- **ME364 Systematic Engineering Design**
- **Meeting day(s) and time(s)** (face-to-face)
  - **Lecture**: MW 9:30-10:20am WALC 1132
  - **Lab**: M 12:30-2:30pm ME1178
- **Credits:** 3 (2 lecture hours + 2 lab hours)
- **Prerequisites:**
  - ME264: Intro. to manufacturing
  - ME270 Basic Mechanics I
  - ME274 Basic Mechanics II
  - ME323: Mechanics of Material

Course Description:

This course introduces students to a structured and methodical approach to engineering design. Topics include problem definition, concept generation, evaluation methods, embodiment design, detail design and prototyping. Students will work in teams on an open-ended design project, applying the systematic design process to develop functional prototypes.

Learning Outcomes:

By the end of the course, students will be able to:

1. Apply a structured design methodology to solve open-ended engineering problems.
2. Utilize engineering principles (statics/dynamics/mechanics of material) to develop and evaluate design concepts.
3. Develop and refine prototypes using CAD, Simulation, and fabrication/manufacturing tools.
4. Work collaboratively in multidisciplinary teams to execute a design project.
5. Communicate design decisions effectively through reports and presentations.

Learning Resources:

1. Gerhard Pahl ,Wolfgang Beitz, Jörg Feldhusen, Karl-Heinrich Grote. Engineering Design, A systematic approach. (Free PDF available online)
2. Karl Ulrich, Steven Eppinger, Product Design and Development
3. Robert L. Norton, Design of Machinery

Instructor(s) Contact Information:

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Assignments:

Posted on BrightSpace

Grading:

- Design Project (Team-based, includes Peer Evaluation): 50%
- Homework & Assignments: 50%

Notes:

- The open-ended **team-based design project** will run throughout the semester, integrating lectures and lab topics.
- Labs will emphasize **hands-on learning and iterative design** to reinforce theoretical concepts.

Attendance Policy:

Refer Brightspace University Policies and Statements

Schedule:

Posted on BrightSpace

Academic Integrity:

Refer Brightspace University Policies and Statements

Nondiscrimination Statement:

https://www.purdue.edu/home/ea_eou_statement/

Accessibility:

https://www.purdue.edu/drc/index.php

Mental Health/Wellness Statement:

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.-5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparedness:

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.

See Purdue's Information on Emergency Preparation and Planning. This website covers topics such as Severe Weather Guidance, Emergency Plans, and a place to sign up for the Emergency Warning Notification System. I encourage you to download and review the *Emergency Preparedness for Classrooms document*.

Please make note of items like:

- The location to where we will proceed after evacuating the building if we hear a fire alarm.
- The location of our Shelter in Place in the event of a tornado warning.
- The location of our Shelter in Place in the event of an active threat such as a shooting.